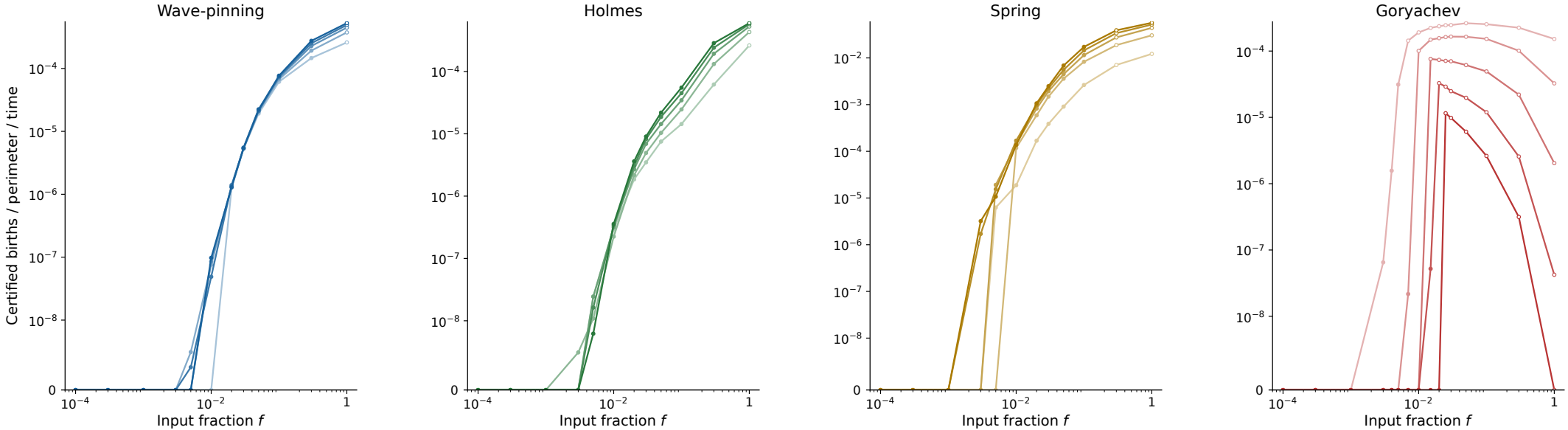
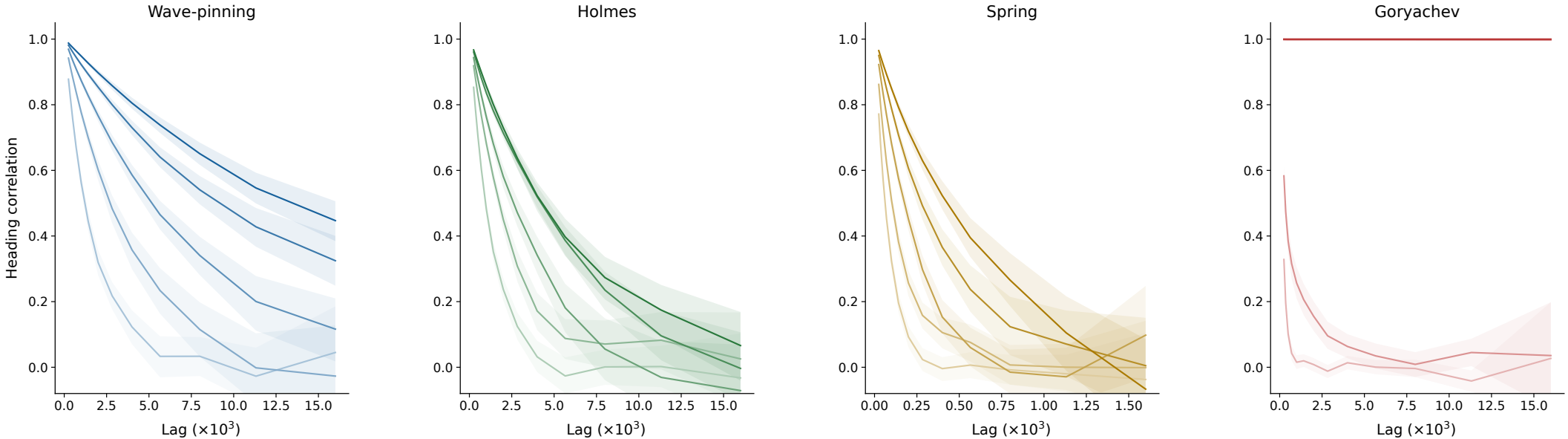
S4.1 Certified persistent-secondary birth rates



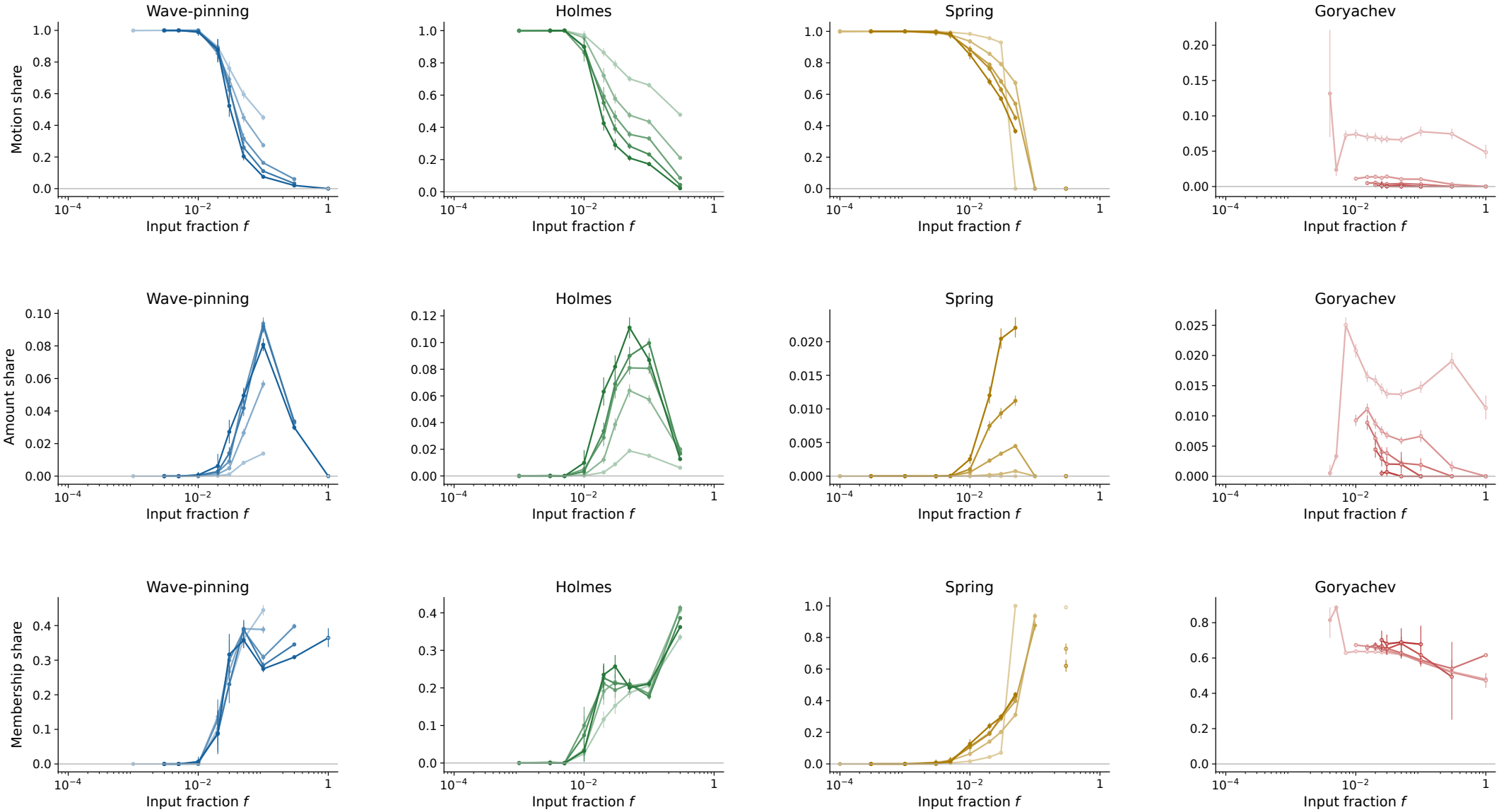
Rates are certified lower bounds per actual perimeter–time. Uncertified onsets can contain split descendants; native local parameters are unchanged.

S4.1b Native heading correlations across all five sizes



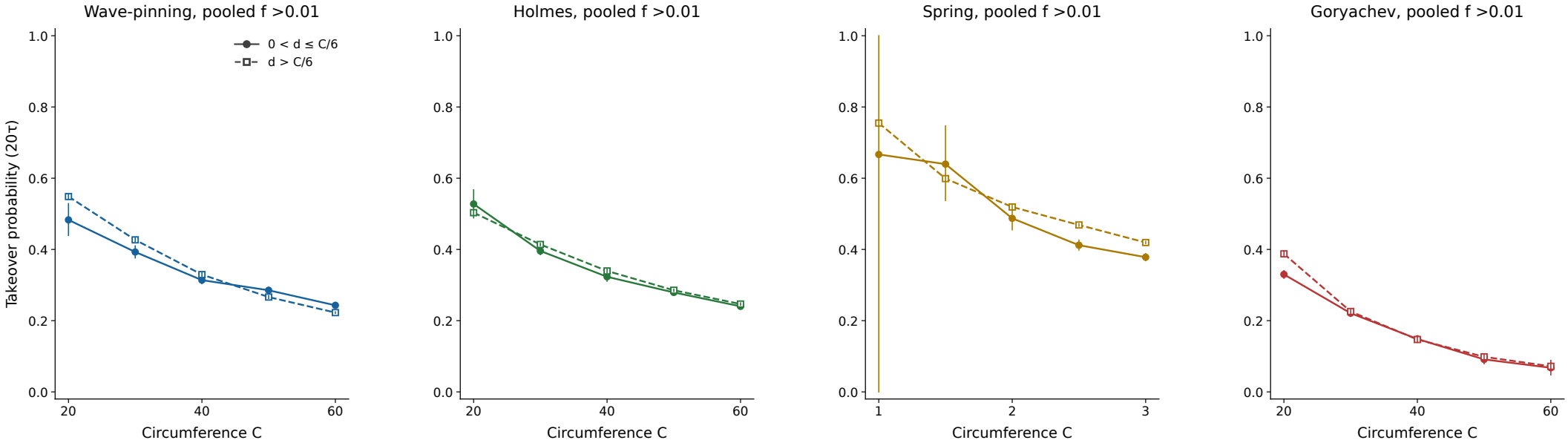
Transition inputs match main C–D. Whole-history 95% intervals; light to dark denotes increasing circumference. Endpoint coherence |P|≥0.1. All points are in condition_lag_metrics.csv.

S4.2 Motion, amount and membership shares



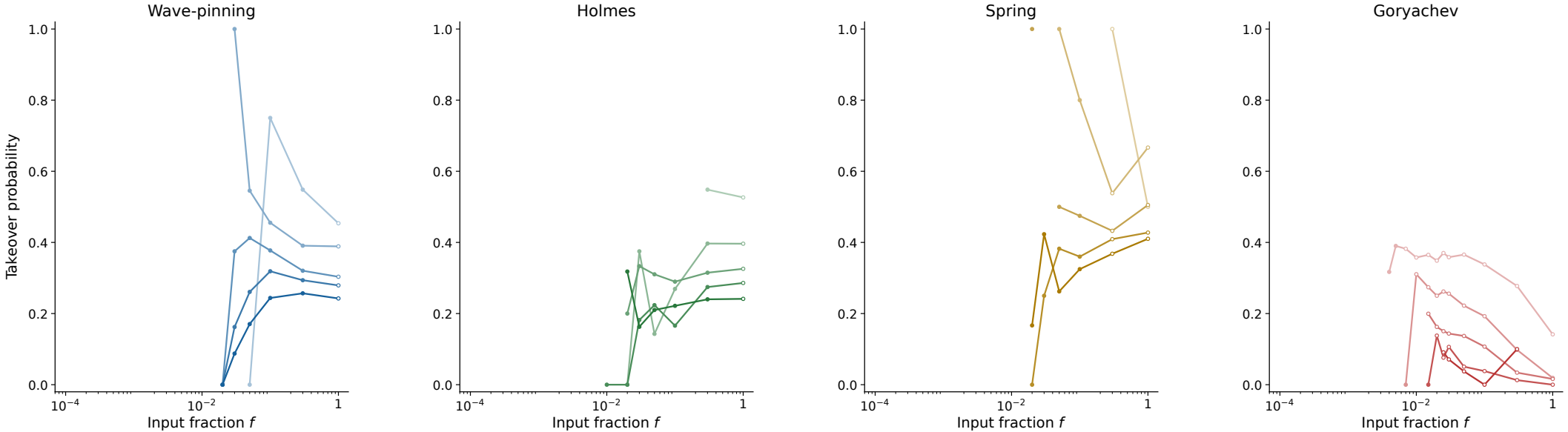
Native-amplitude symmetric Shapley convention. Signed fractions can be negative or exceed one; Eligibility requires coherent native heading at both endpoints.

S4.3 Pooled finite-horizon takeovers



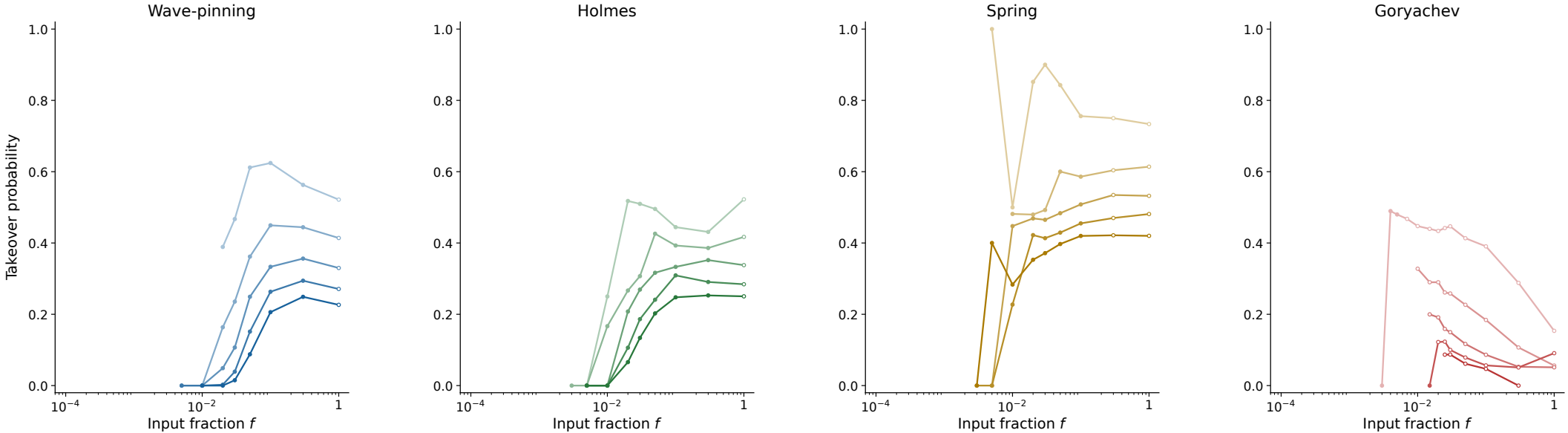
Pooled f > 0.01 diagnostic, shown only for positive distance. Main J uses a separately selected input per model and >=200 births at every size.

S4.4 Takeover probability at every input: 0 < d ≤ C/6



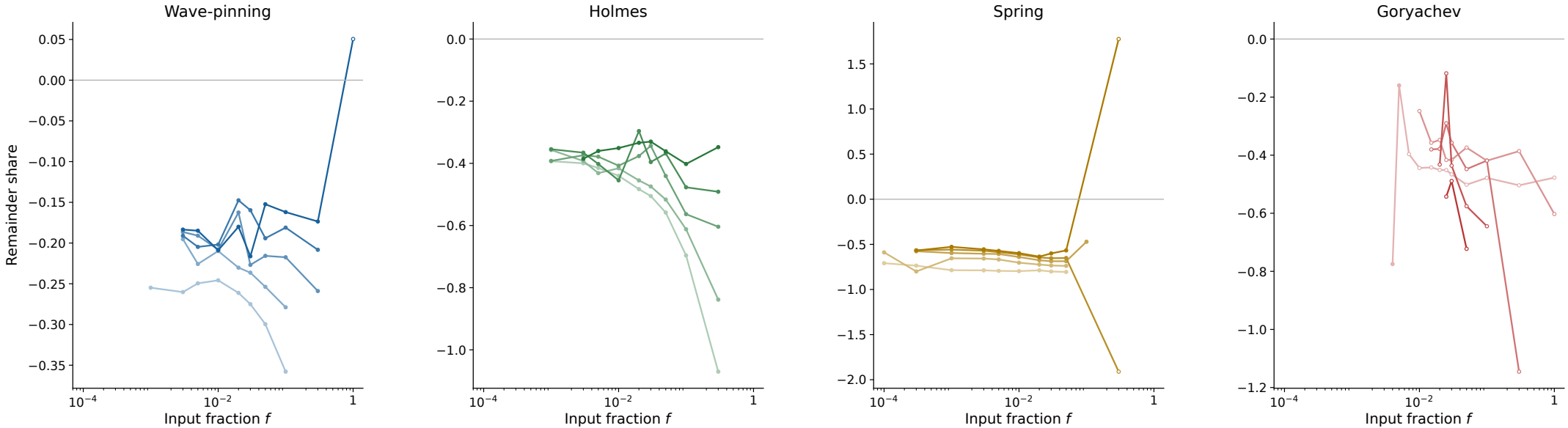
Common follow-up 20τ; each curve is one size. Certified no-overlap subset only. Inputs with no certified births remain missing.

S4.4 Takeover probability at every input: d > C/6



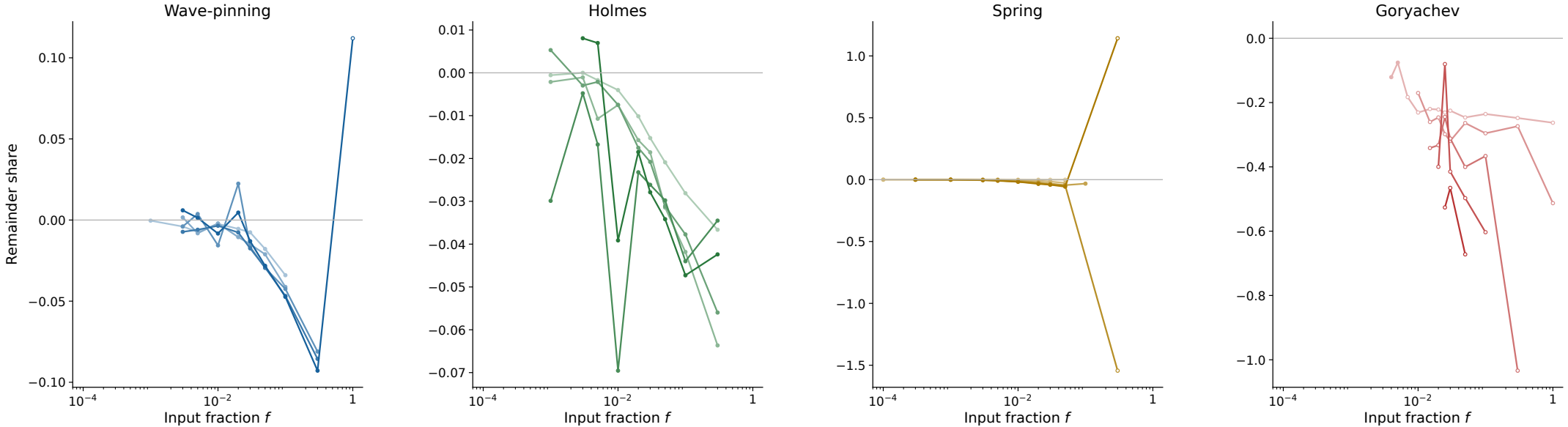
Common follow-up 20τ; each curve is one size. Certified no-overlap subset only. Inputs with no certified births remain missing.

S4.5 Own-mask moment check: point matched mass



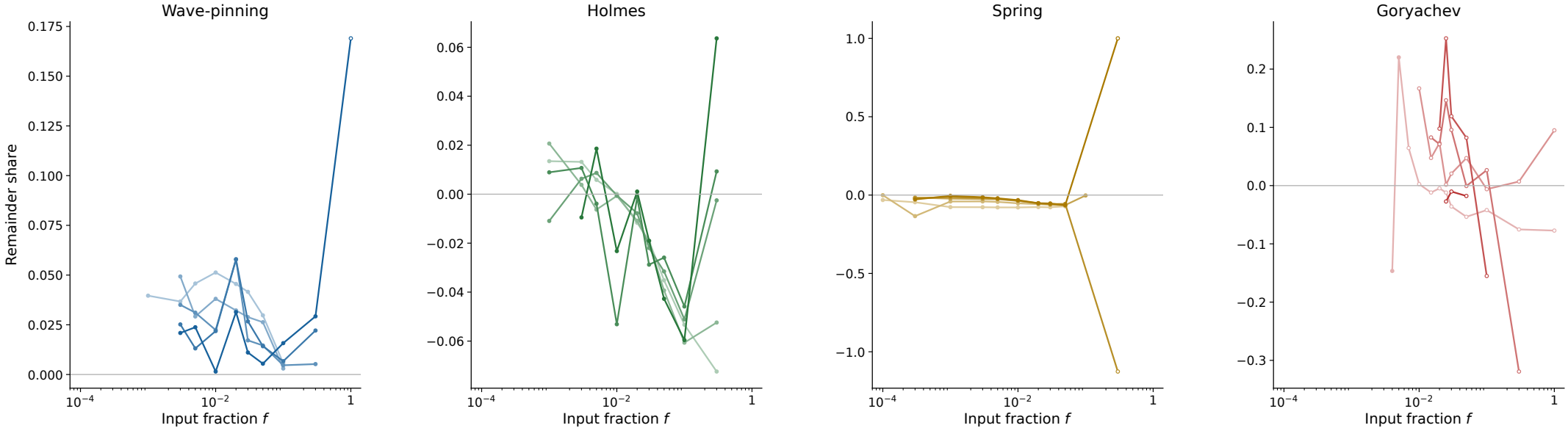
Existing seeds 0/1; saved binned float16 profiles. Native-mass denominator in first 3 variants; last 2 use region-total normalization and native amplitude scaling. Shape enters amount/shape in first 3 and motion/shape in last 2. Algebraic reconstruction is not causal validation.

S4.5 Own-mask moment check: exact full mask



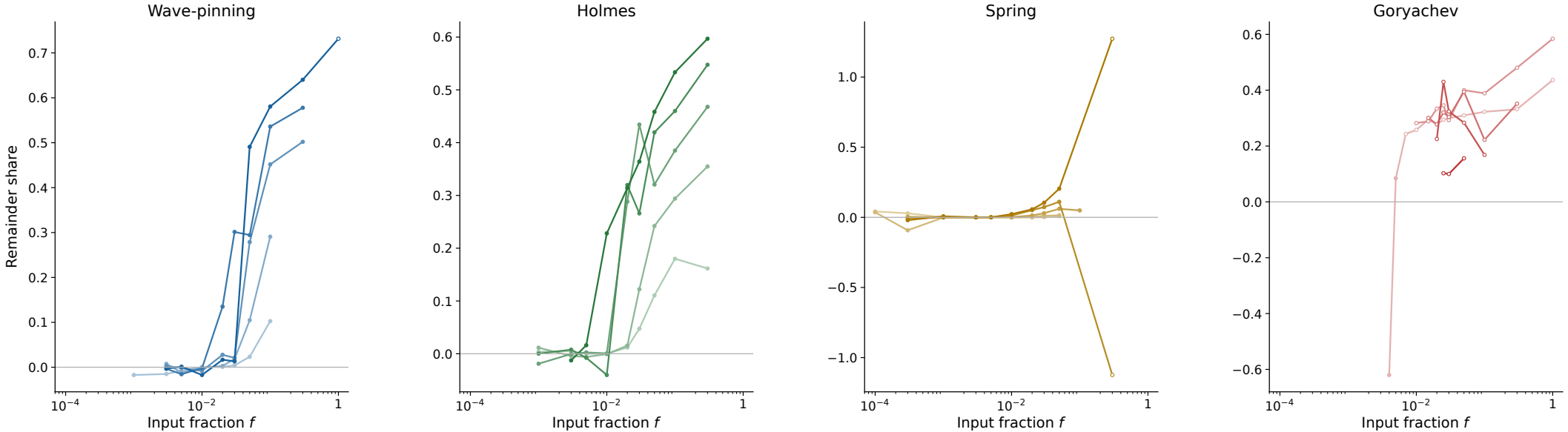
Existing seeds 0/1; saved binned float16 profiles. Native-mass denominator in first 3 variants; last 2 use region-total normalization and native amplitude scaling. Shape enters amount/shape in first 3 and motion/shape in last 2. Algebraic reconstruction is not causal validation.

S4.5 Own-mask moment check: exact contour mask



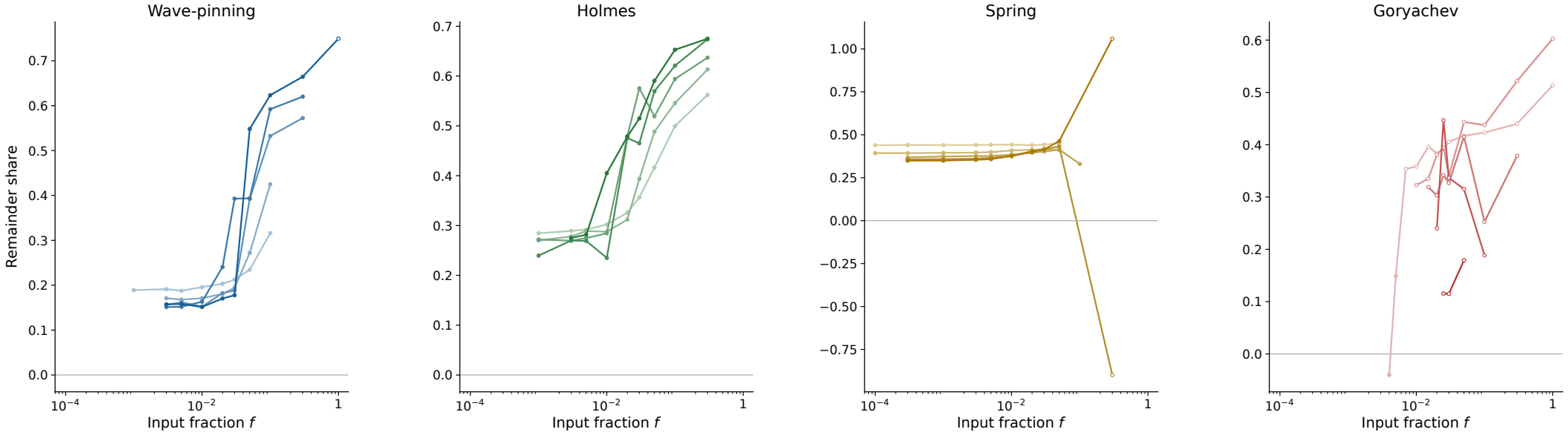
Existing seeds 0/1; saved binned float16 profiles. Native-mass denominator in first 3 variants; last 2 use region-total normalization and native amplitude scaling. Shape enters amount/shape in first 3 and motion/shape in last 2. Algebraic reconstruction is not causal validation.

S4.5 Own-mask moment check: round5 point native amplitude



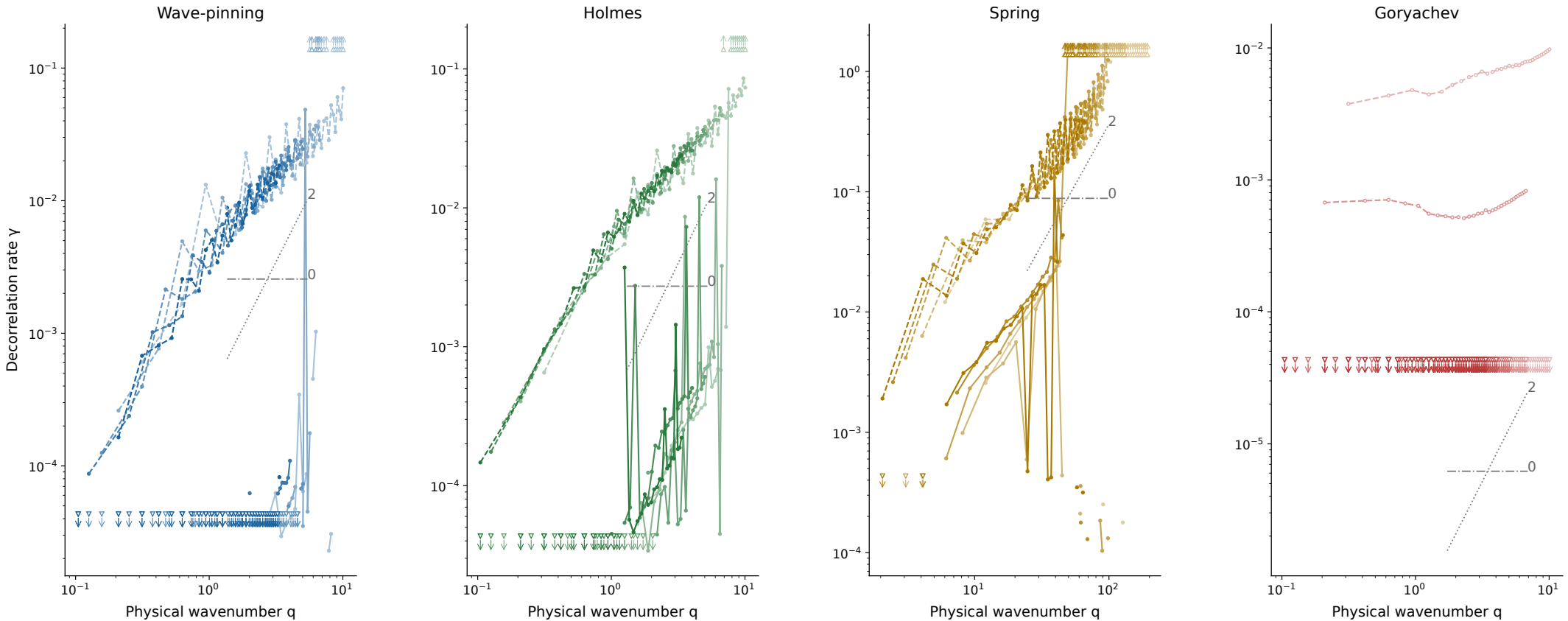
Existing seeds 0/1; saved binned float16 profiles. Native-mass denominator in first 3 variants; last 2 use region-total normalization and native amplitude scaling. Shape enters amount/shape in first 3 and motion/shape in last 2. Algebraic reconstruction is not causal validation.

S4.5 Own-mask moment check: round5 exact native amplitude



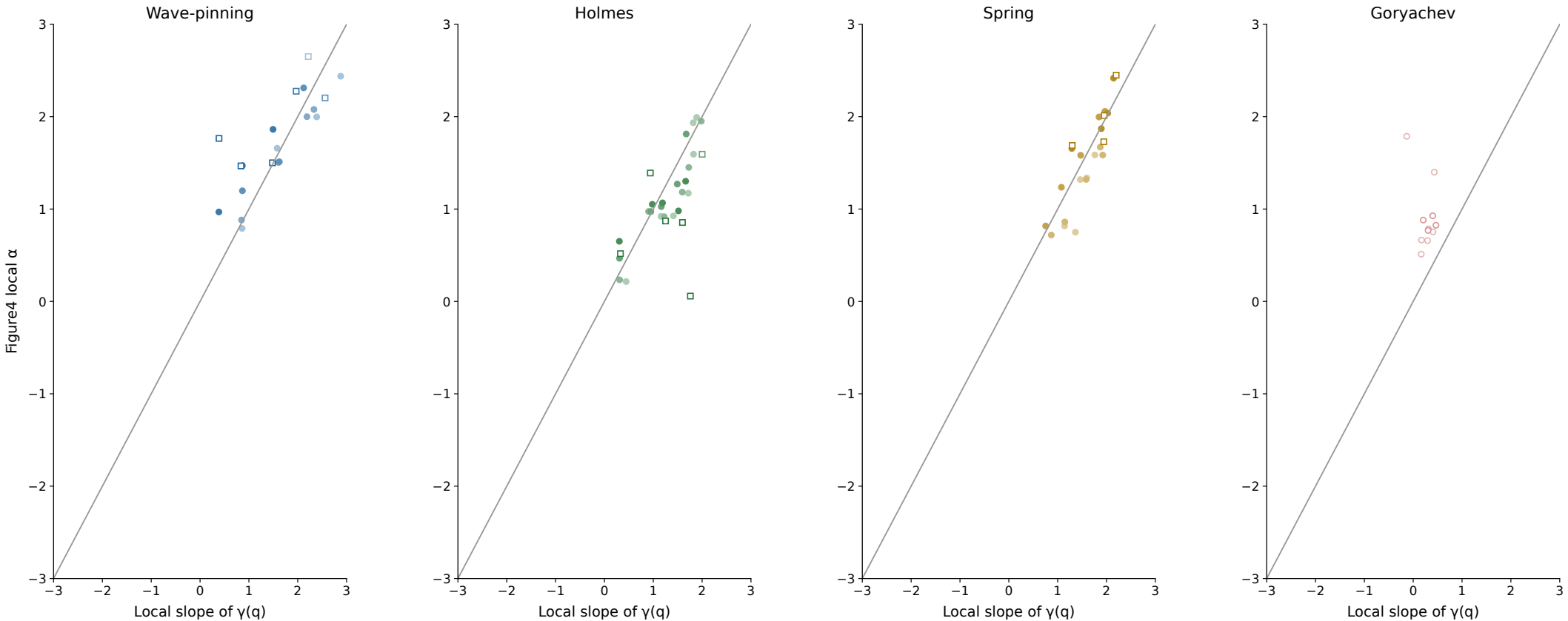
Existing seeds 0/1; saved binned float16 profiles. Native-mass denominator in first 3 variants; last 2 use region-total normalization and native amplitude scaling. Shape enters amount/shape in first 3 and motion/shape in last 2. Algebraic reconstruction is not causal validation.

K Fourier-mode decorrelation versus physical wavenumber



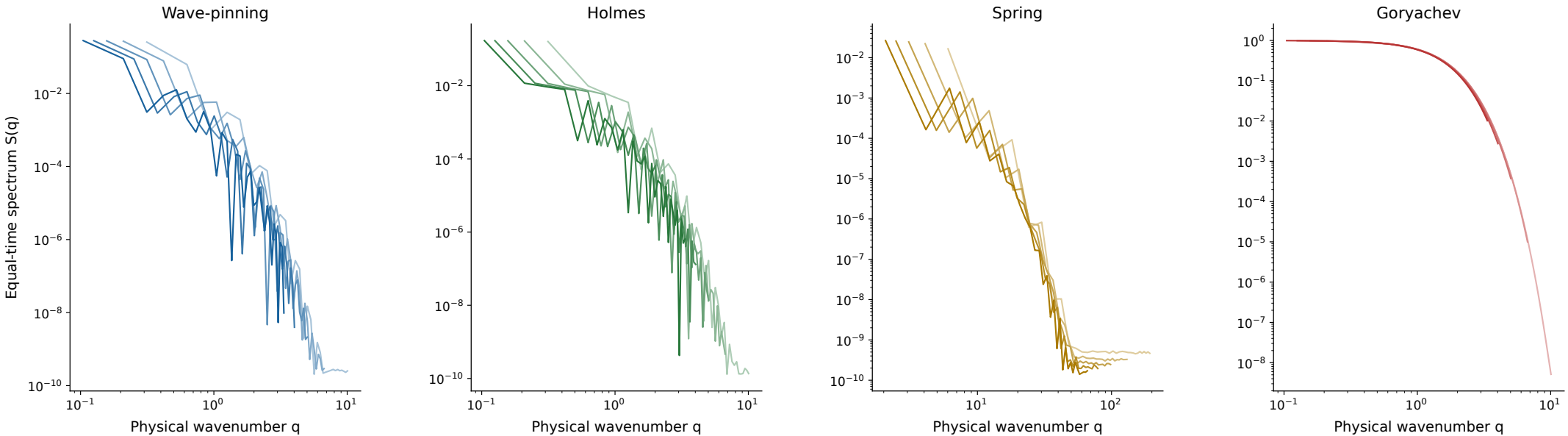
Solid weak f = 0.001 (existing exemplars); dashed transition input. Open triangles show inverse-half-time window bounds, conditional on a monotone first crossing. Strong curves use the independent mode-only cohort when available; source and resolution flags are exported.

L Size exponent versus independently predicted local mode slope



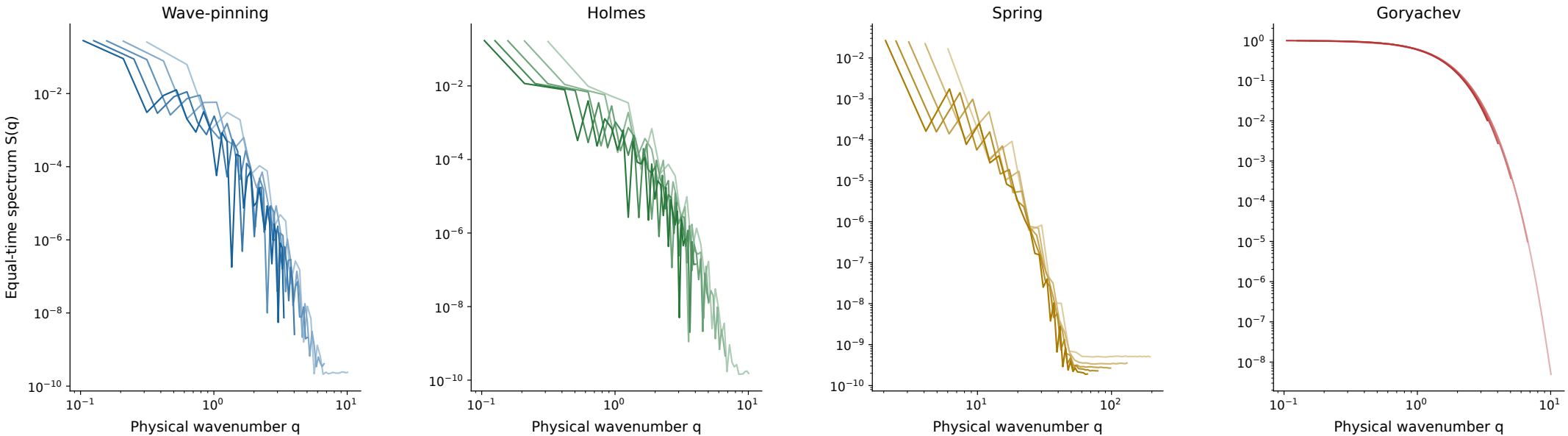
Mode slope uses the nearest 8 mode points from other sizes at the same model and input; the target size is excluded. One-sided predictions are open squares. Forced conditions have open circles. This tests collapse rather than replotting the same size slope.

S4.6 Equal-time spectrum, f = 0.0001



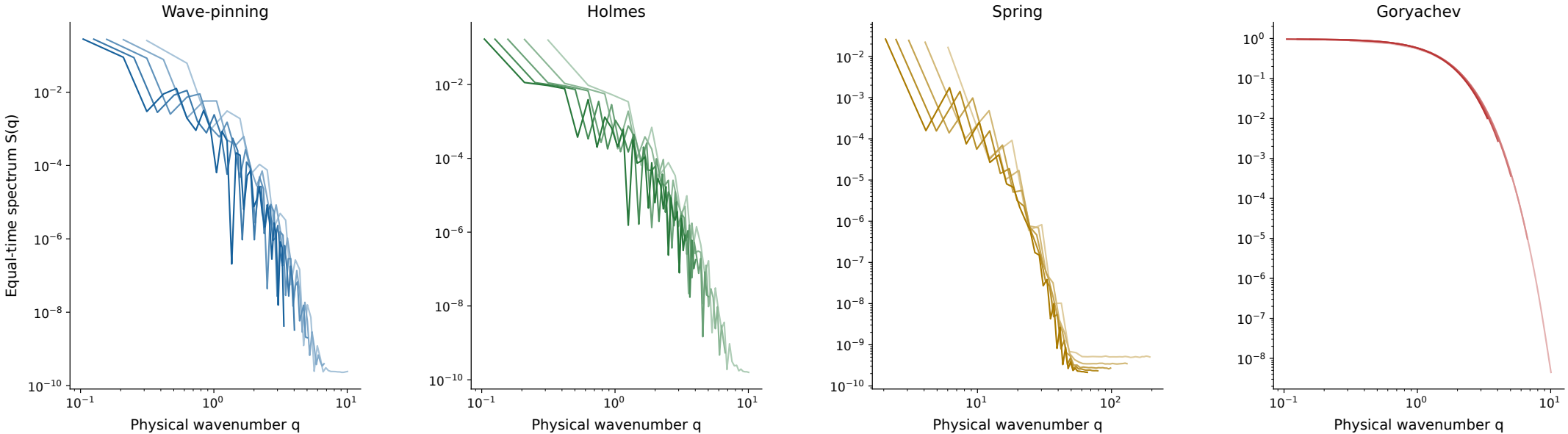
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.0003



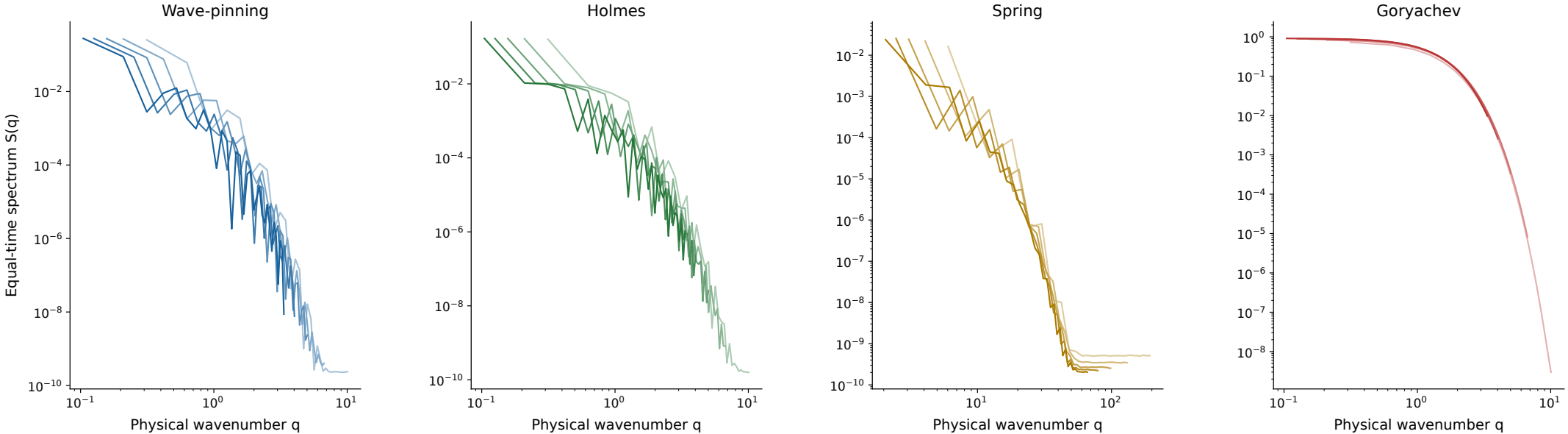
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.001



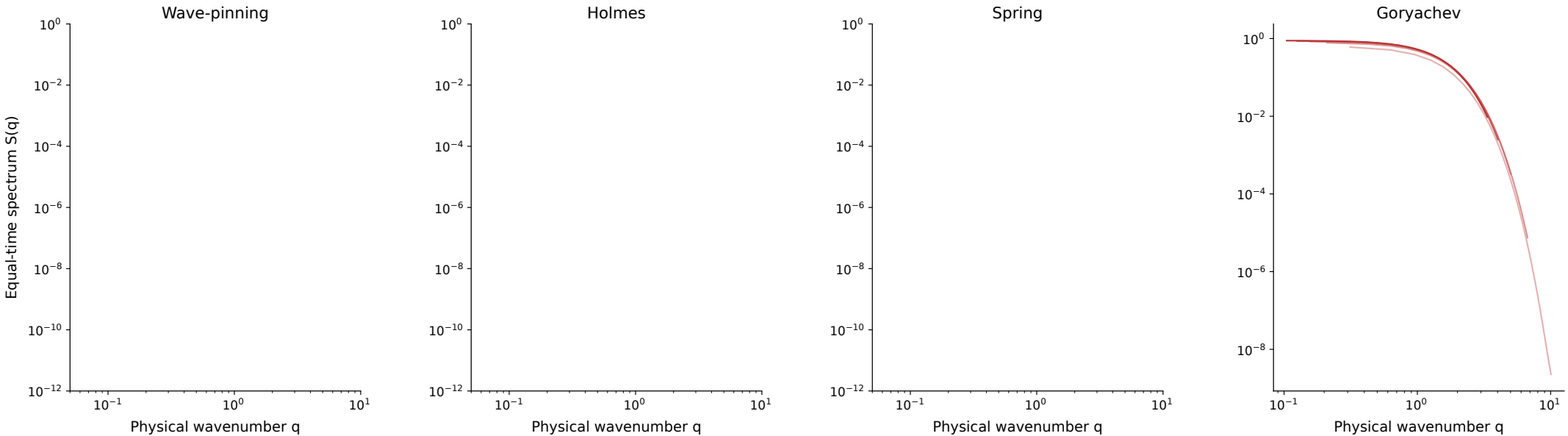
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.003



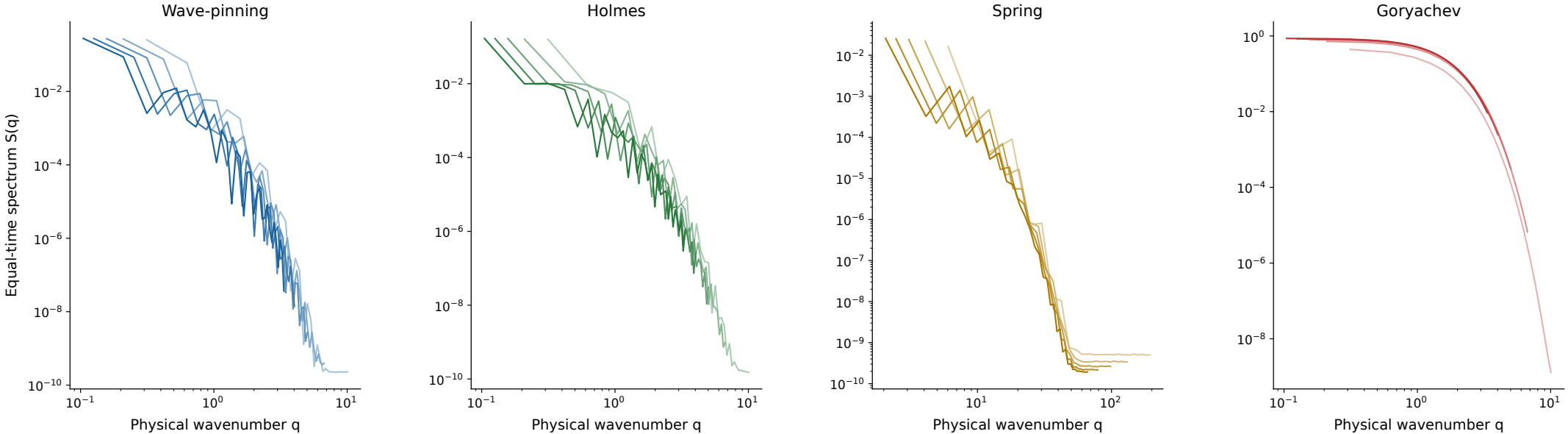
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.004



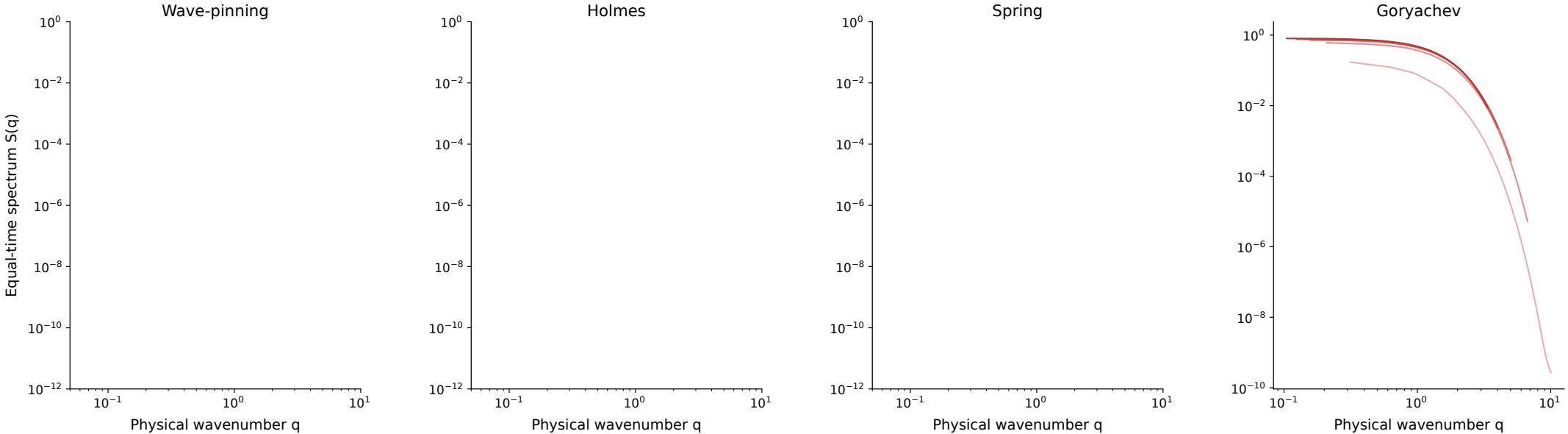
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.005



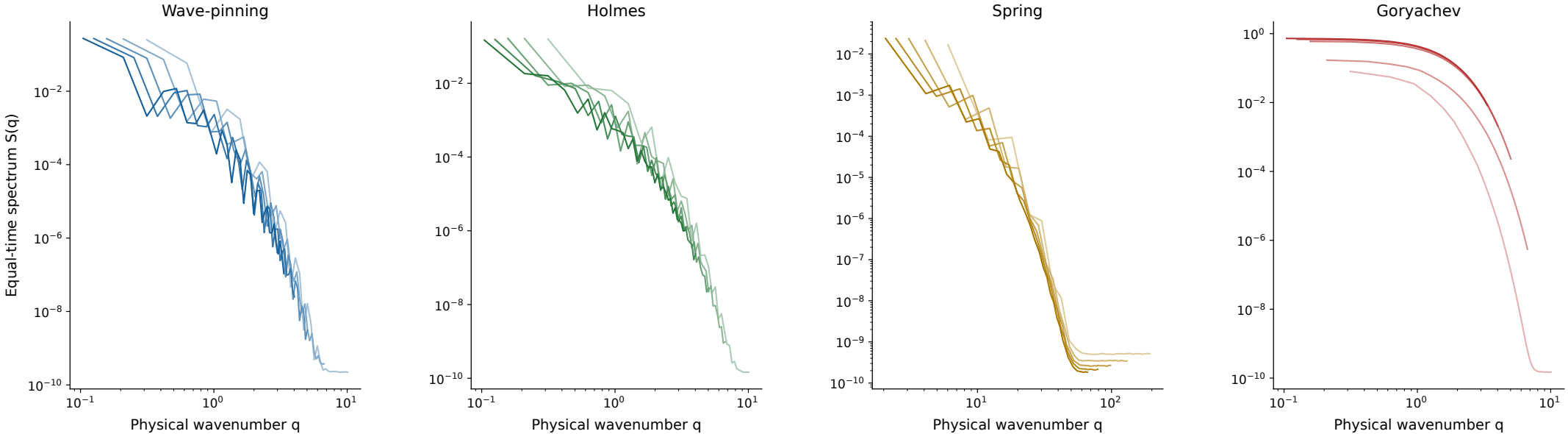
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.007



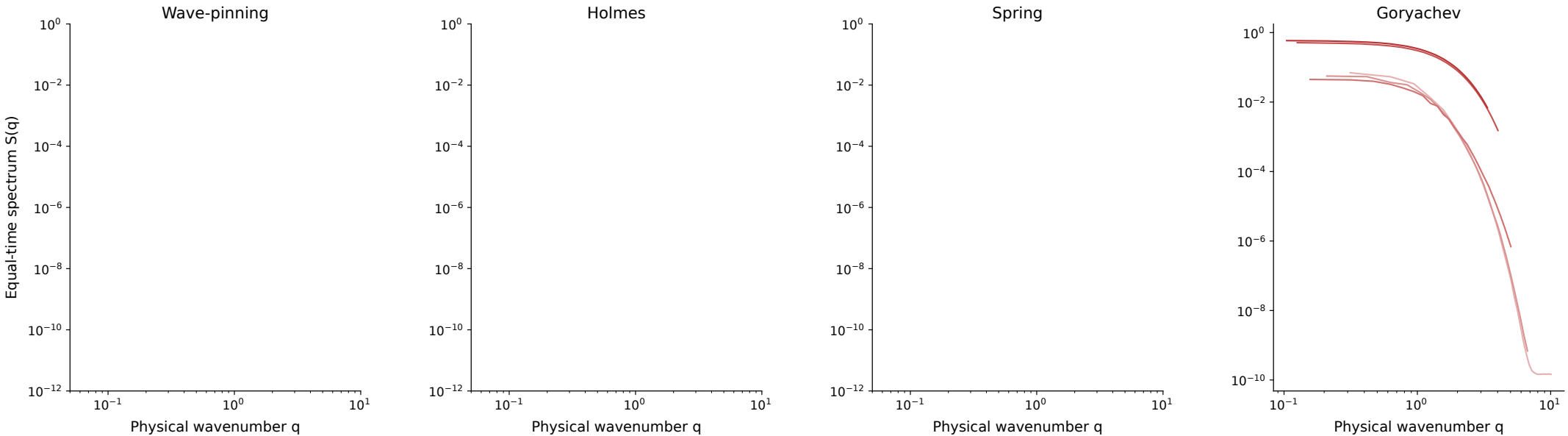
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.01



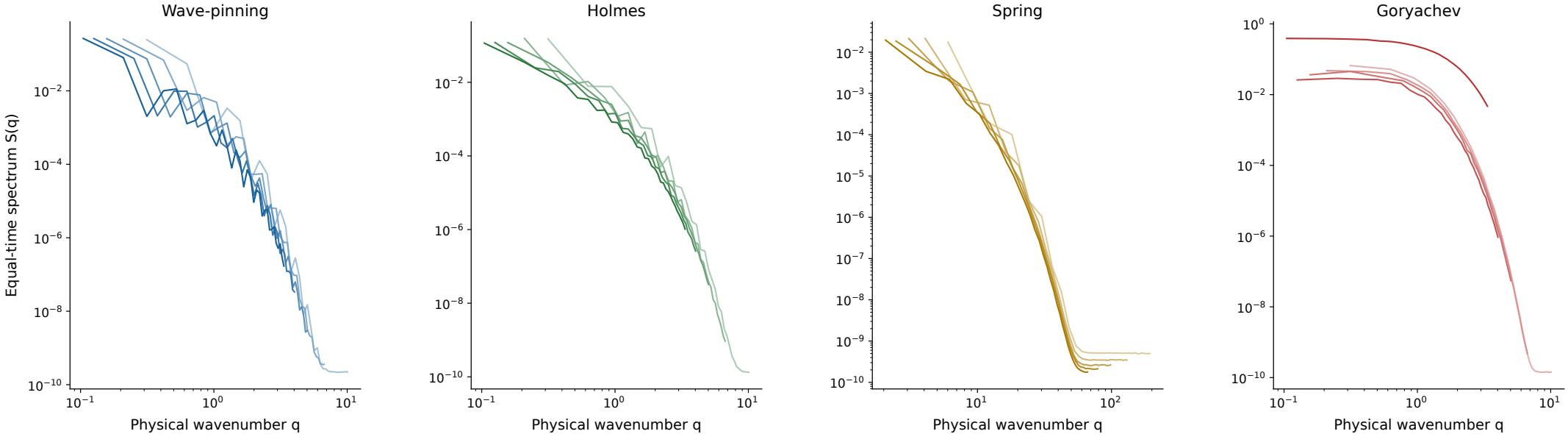
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.015



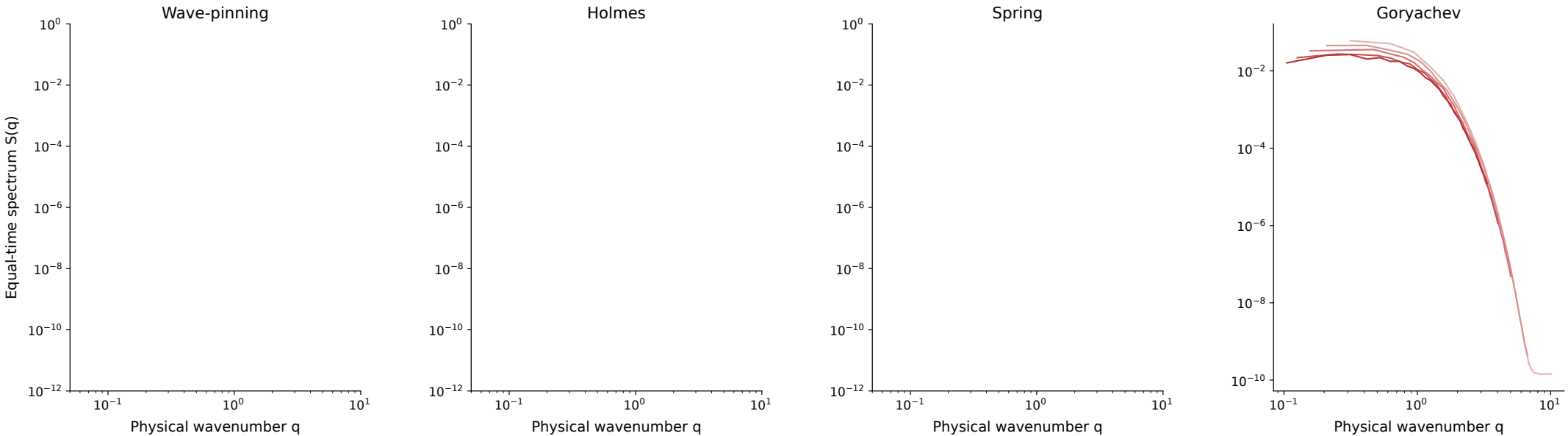
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.02



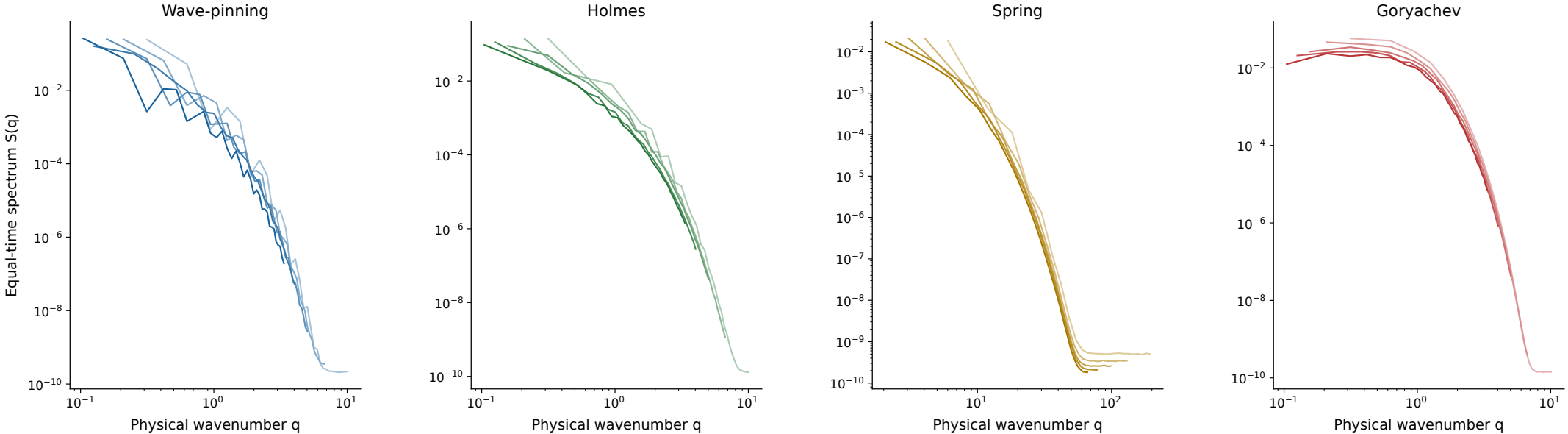
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.025



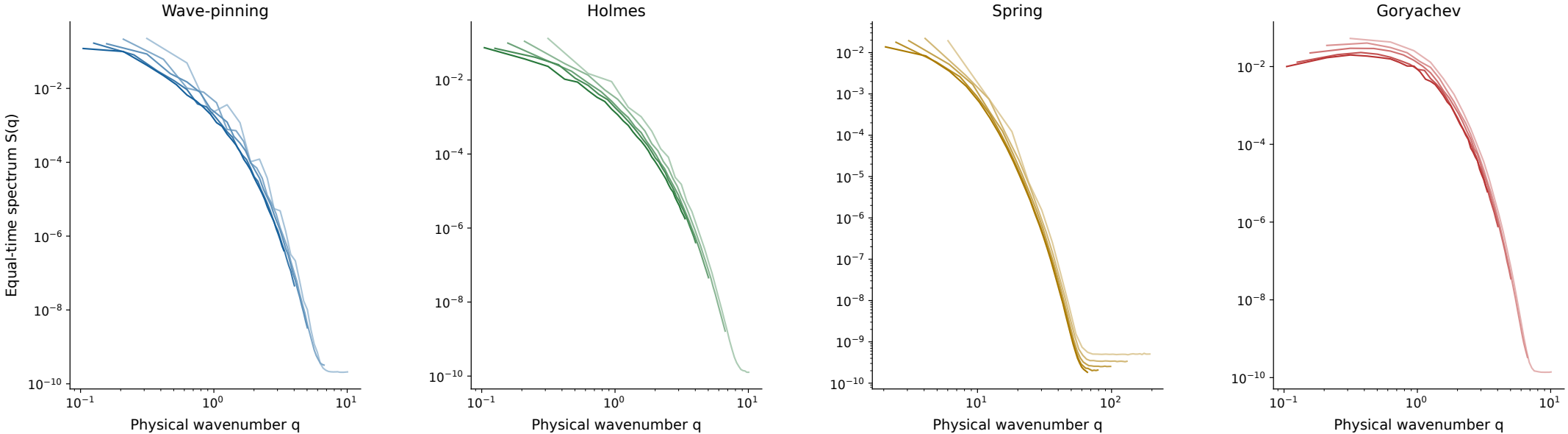
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.03



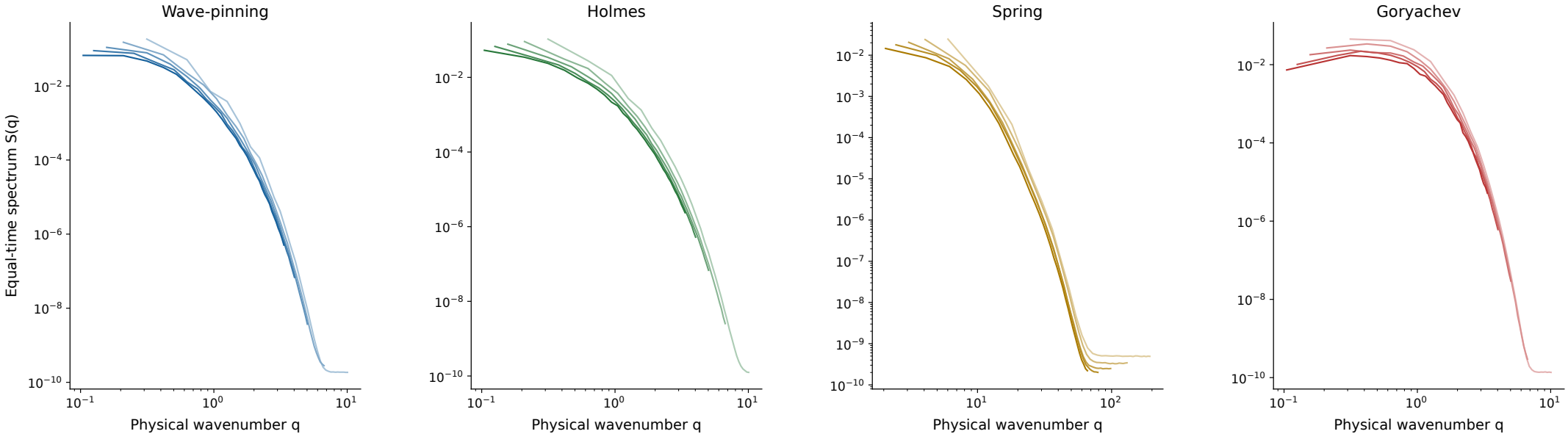
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.05



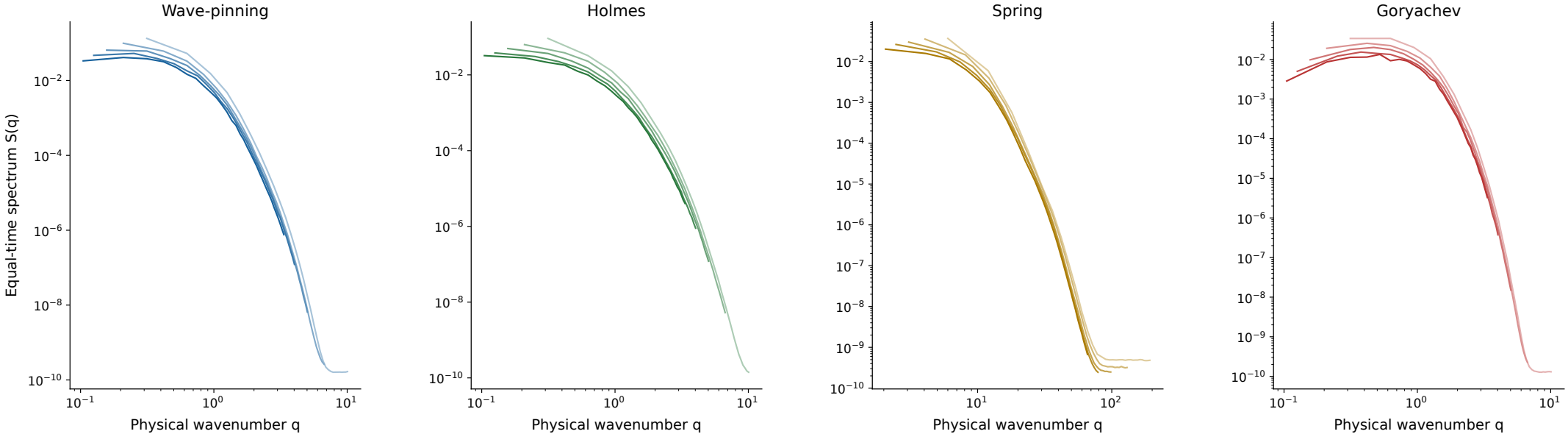
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.1



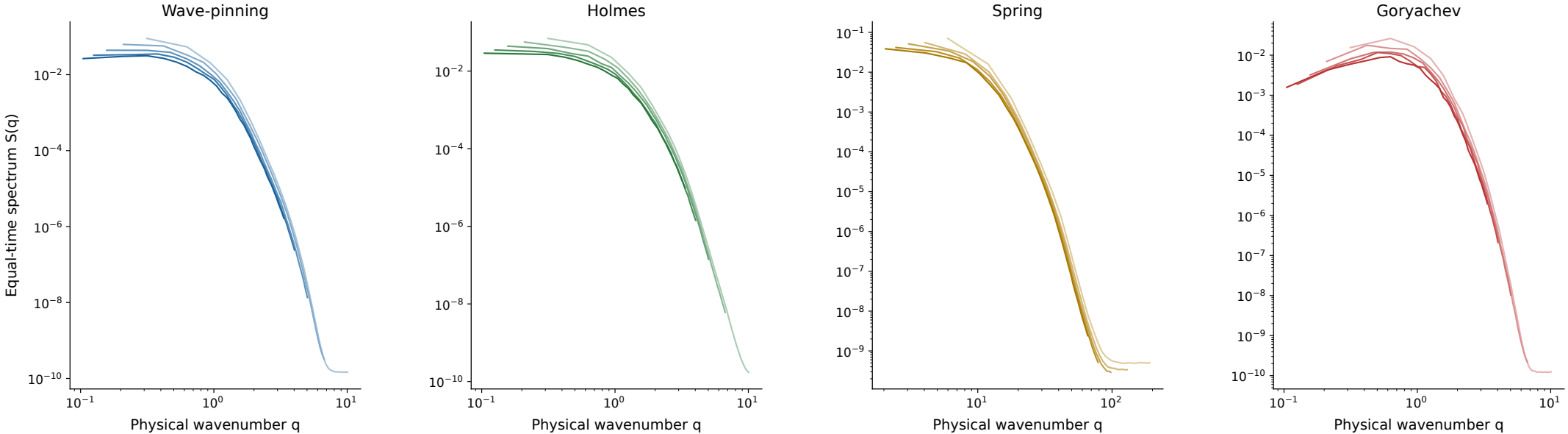
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 0.3



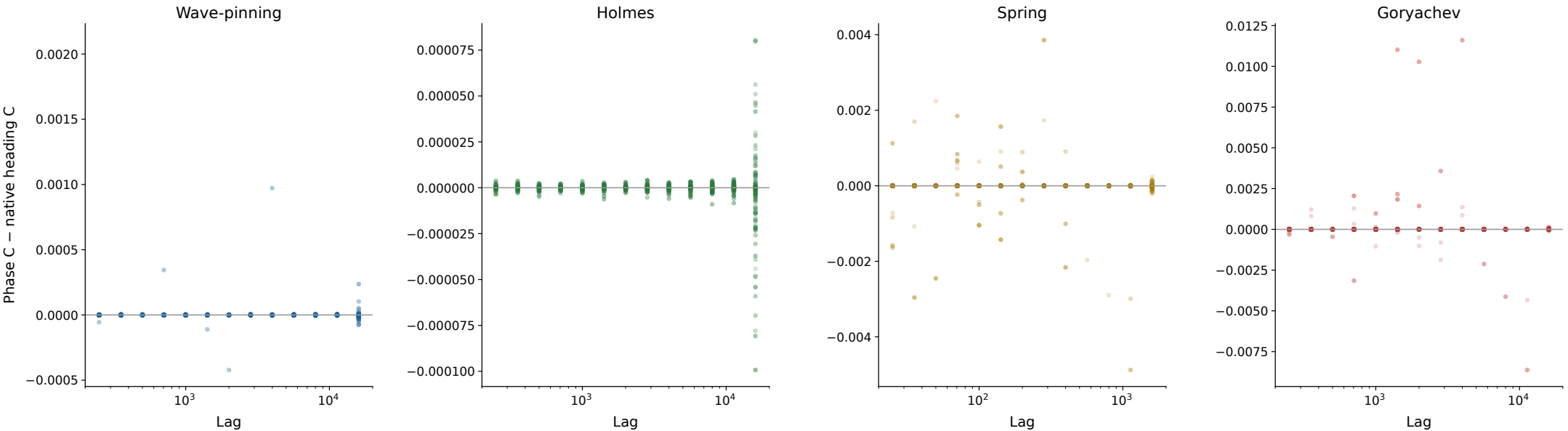
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.6 Equal-time spectrum, f = 1



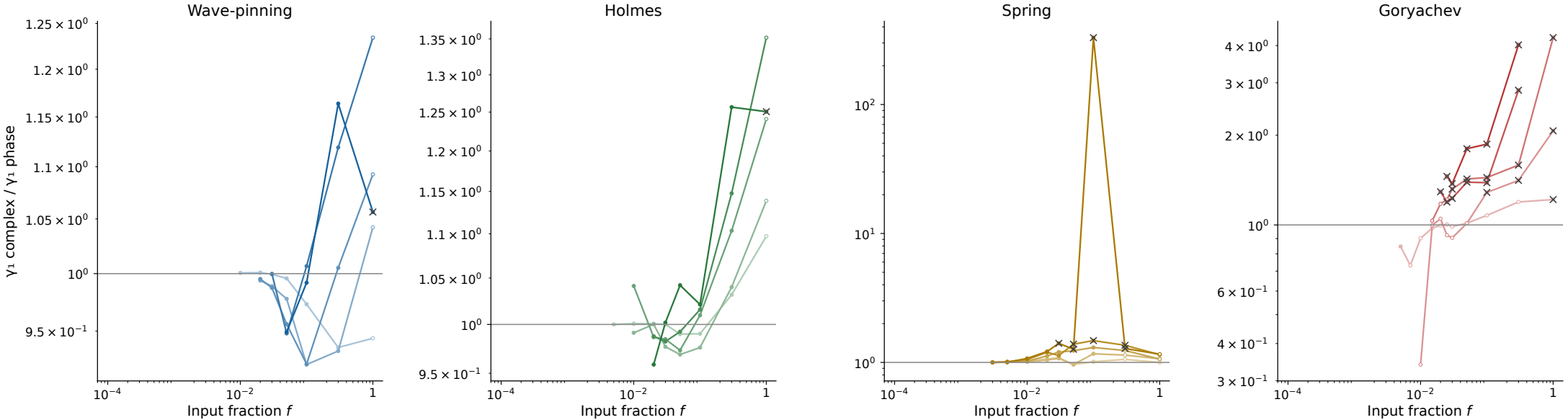
$S(q) = \langle |a_n|^2 \rangle / \langle a_0^2 \rangle$; n=1–32, two existing exemplars/condition. Spring q uses mean C. n* and C/n* are in exemplar_rates.csv.

S4.7 First-mode phase reproduces native heading correlation



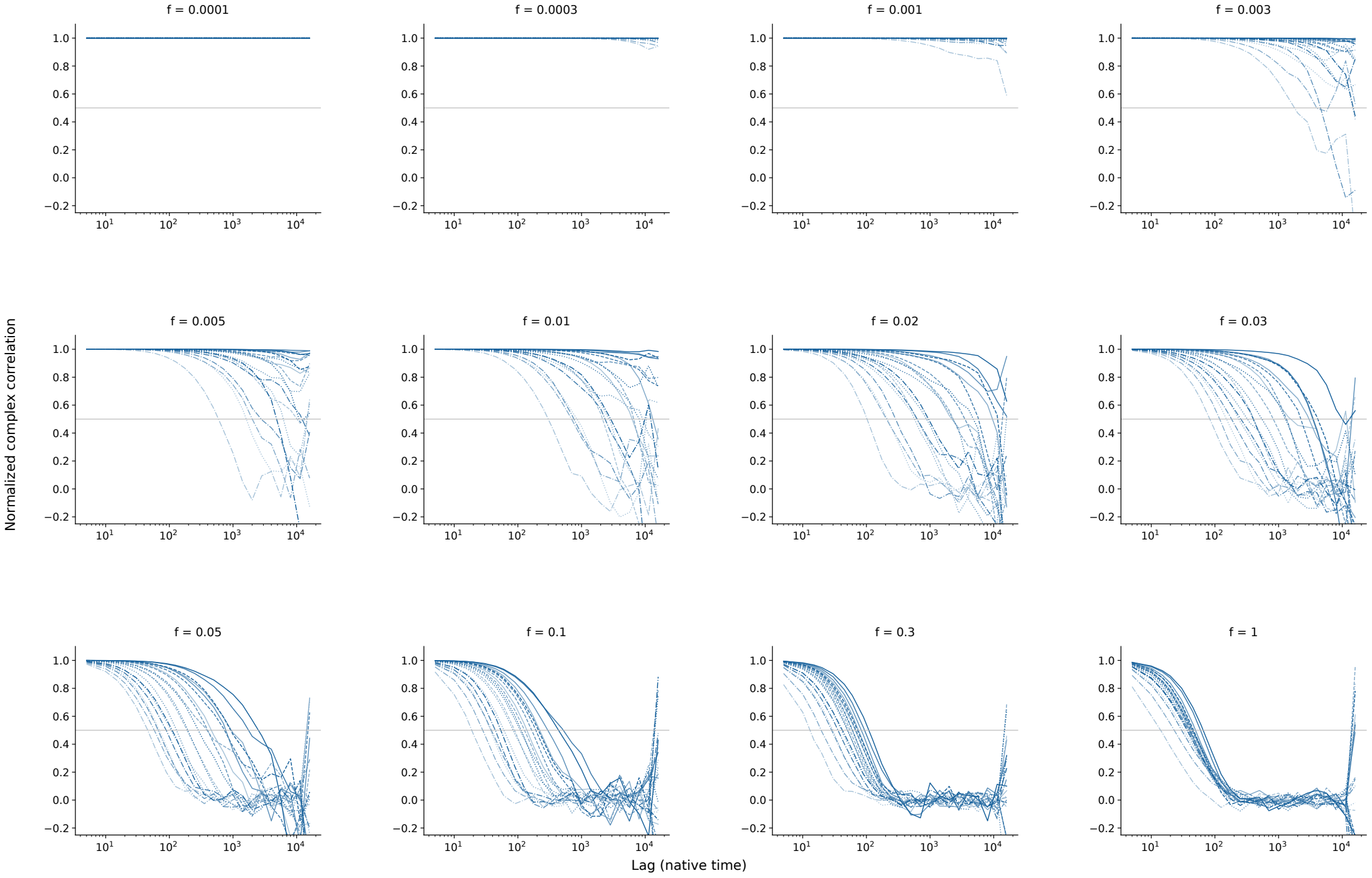
Difference between Fourier a1 phase correlation and native compact P heading, at identical exact saved endpoints. Eligibility uses each stored |P|≥0.1; nonzero differences include threshold crossing and bin/float16 precision.

S4.7b Complex-mode and phase-only first-mode rates



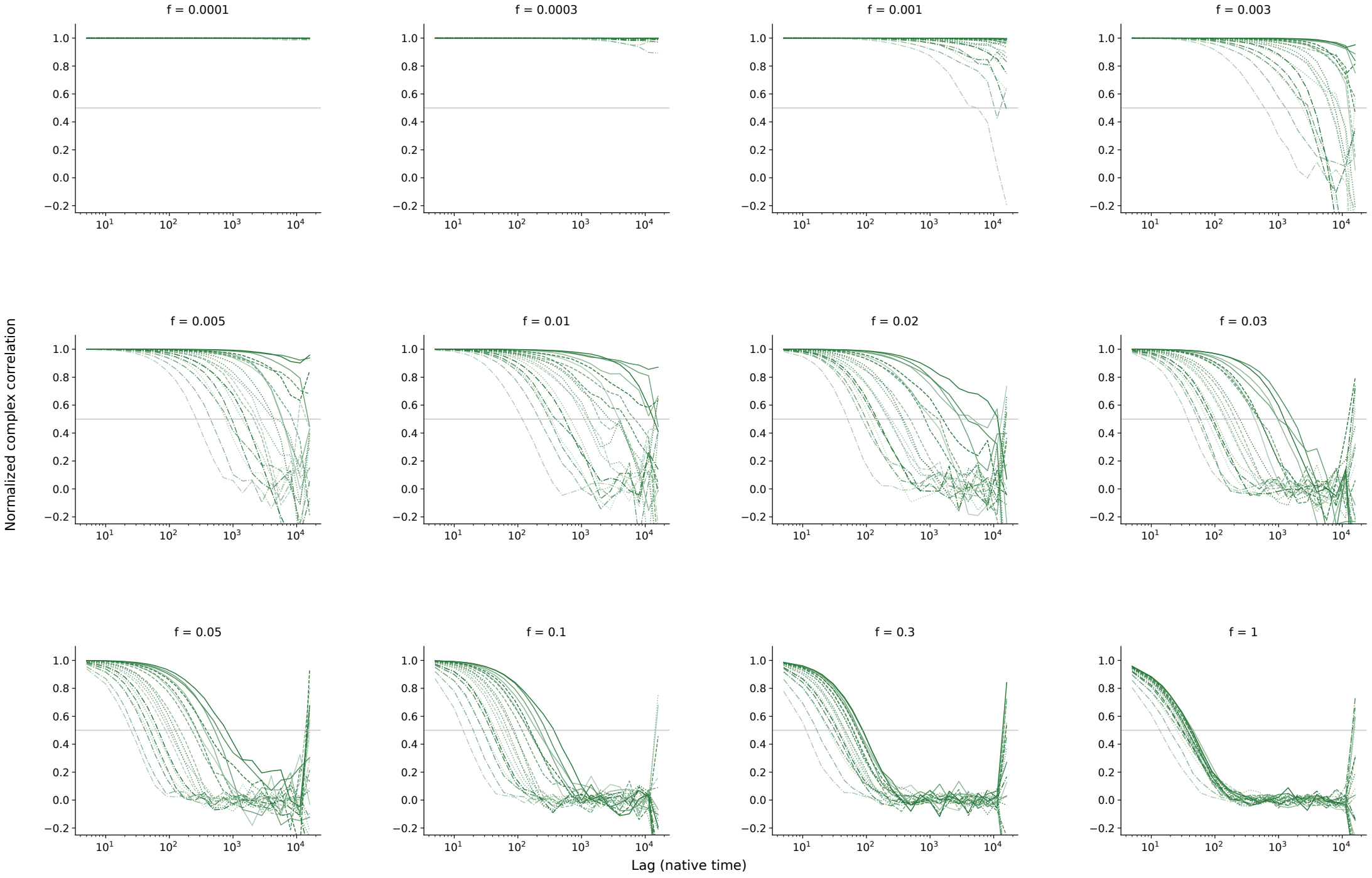
Same cohort, nearest-half lag rule. Complex: amplitude-weighted all endpoints; phase: |P|≥0.1, point estimates without bootstrap precision. Crosses flag phase coverage <0.5. The extreme Spring ratio uses only 4 eligible terminal-lag pairs; all point estimates remain exported.

S4.8 Normalized complex autocorrelations: Wave-pinning



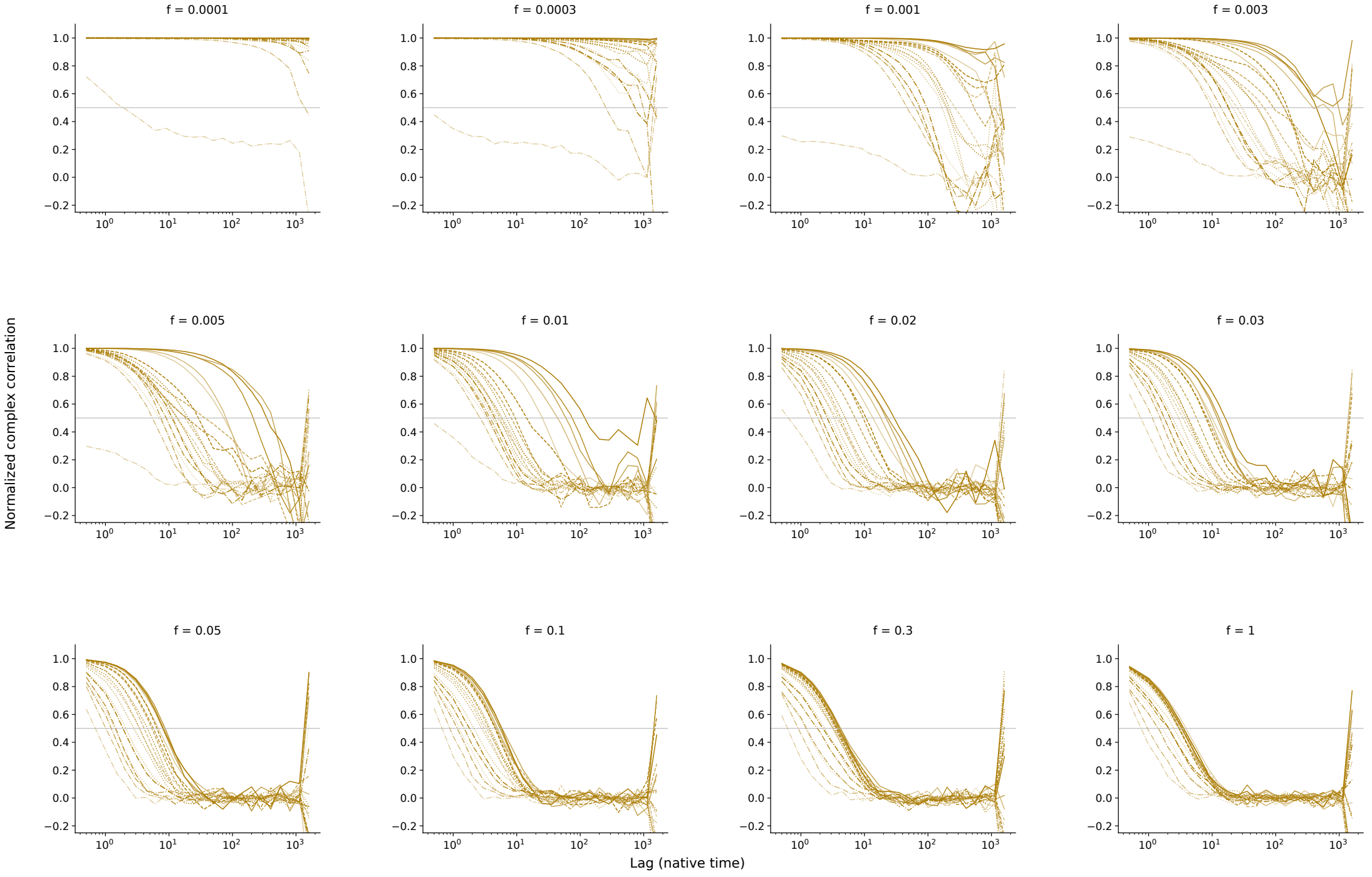
n = 1 solid, 2 dashed, 4 dotted, 8 dash-dot; size light→dark. All 33 modes including n = 0 are retained in compact CSV.

S4.8 Normalized complex autocorrelations: Holmes



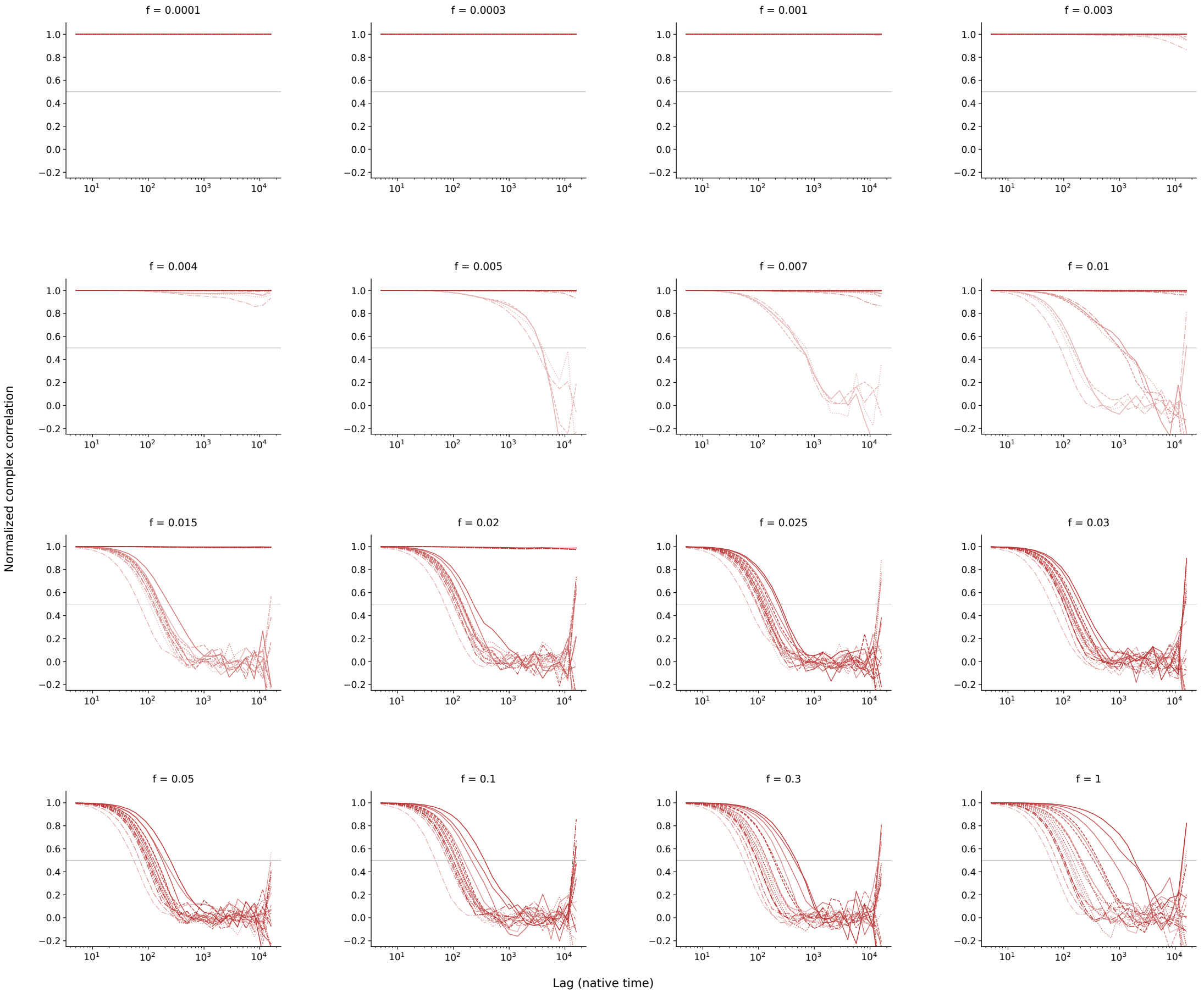
n = 1 solid, 2 dashed, 4 dotted, 8 dash-dot; size light→dark. All 33 modes including n = 0 are retained in compact CSV.

S4.8 Normalized complex autocorrelations: Spring



n = 1 solid, 2 dashed, 4 dotted, 8 dash-dot; size light→dark. All 33 modes including n = 0 are retained in compact CSV.

S4.8 Normalized complex autocorrelations: Goryachev



n = 1 solid, 2 dashed, 4 dotted, 8 dash-dot; size light→dark. All 33 modes including n = 0 are retained in compact CSV.